Two-Scale Reynolds Decomposition of VCML: The Interface Formula, Inheritance Roughness, and the Two Effective PDEs

Gabriel Aubin-Moreau

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Abstract

Papers 20–22 established a spatial picture of VCML: a three-equation ODE (Paper 20), a birth-mediated C-factor (Paper 21), and an interface-sharpening signature in the within-zone correlation length ξ (Paper 22). Here we derive the *two-scale Reynolds decomposition* of VCML, splitting fieldM into zone means and within-zone residuals: $F(x,t) = \bar{F}_z(t) + F_{\text{res}}(x,t)$. This yields two coupled effective PDEs. PDE 1 (zone-mean scale, Papers 20–21) governs when zones differentiate, with timescale $\tau_{\text{buildup}} \propto \nu^{-0.40}$. PDE 2 (within-zone scale) predicts an equilibrium interface width $\xi_\infty = C_{\text{geo}}\sqrt{\kappa/(P_{\text{consol}} \cdot FA)}$, where the within-zone effective diffusivity is κ (not D_{copy}). We test both predictions with a κ-sweep and an FA-sweep. The κ-sweep confirms the formula direction ($\xi_\infty \propto \kappa^{+0.28}$, $R^2 = 0.69$; predicted $+0.5$) and a geometric prefactor $C_{\text{geo}} \approx 5.6$. The FA-sweep reveals an unexpected *positive* scaling ($\xi_\infty \propto FA^{+0.21}$, $R^2 = 0.79$; predicted -0.5), identifying a second mechanism: birth-mediated inheritance roughness. High FA drives F toward larger-amplitude zone-specific values; newborn cells inherit a fraction SEED_BETA of this amplitude from their neighbours, creating within-zone roughness that increases with FA. This inheritance mechanism competes with and slightly dominates the interface-sharpening formula in the parameter range tested. The flat ν response ($\xi_\infty \propto \nu^{+0.04}$, from Paper 22) confirms that $D_{\text{eff,within}} = \kappa$, not $D_{\text{copy}}(\nu)$: the two spatial diffusivities operate at different scales.

1 Introduction

The PDE of VCML derived in Paper 21 treats the full fieldM $F(x,t)$:

$$\frac{\partial F}{\partial t} = P_c(x,t) \cdot FA \cdot (m - F) + D_{\text{eff}}\, \nabla^2 F, \tag{1}$$

where $D_{\text{eff}} = \kappa + D_{\text{copy}}(\nu)$ and $D_{\text{copy}} \gg \kappa$.

This single-scale PDE hides a natural two-scale structure. The fieldM contains two components with different characteristic lengths and timescales:

1. **Zone-mean** $\bar{F}_z(t)$: the mean fieldM in zone z, governed by the birth-mediated propagation speed (Paper 21, $\tau_{\text{buildup}} \propto \nu^{-0.40}$).
2. **Within-zone residual** $F_{\text{res}}(x,t) = F(x,t) - \bar{F}_z(t)$: local deviations set by the balance of site-level diffusion κ and consolidation-driven damping, measured in Paper 22 ($\xi_\infty \approx$ 2–3 sites).

Paper 23 derives the two effective PDEs governing each component and tests the key prediction of PDE 2: $\xi_\infty = C_{\text{geo}}\sqrt{\kappa/(P_{\text{consol}} \cdot FA)}$.

2 The Two-Scale Decomposition

2.1 Reynolds decomposition

Following the Reynolds decomposition from turbulence theory, write:

$$F(x,t) = \bar{F}_z(t) + F_{\text{res}}(x,t), \quad \text{where } \bar{F}_z = \frac{1}{w}\int_{\text{zone z}} F\,dx, \tag{2}$$

w = zone width = 5 sites. Substituting into (1) and averaging over zone z gives **PDE 1** (zone-mean dynamics):

$$\frac{d\bar{F}_z}{dt} = \bar{P}_{c,z}\cdot FA\cdot(\bar{m}_z - \bar{F}_z) + \frac{D_{\text{eff}}}{w^2}\left(\bar{F}_{z+1} - 2\bar{F}_z + \bar{F}_{z-1}\right). \tag{PDE 1}$$

The effective zone-level diffusivity is D_{eff}/w^2; this is dominated by $D_{\text{copy}}(\nu)$ (Paper 21: $C \propto \nu^{-0.40}$, κ-independent).

Subtracting the zone average from (1) gives **PDE 2** (within-zone residual dynamics):

$$\frac{\partial F_{\text{res}}}{\partial t} = -P_c\cdot FA\cdot F_{\text{res}} + \kappa\,\frac{\partial^2 F_{\text{res}}}{\partial x^2} + \underbrace{P_c\cdot FA\cdot m_{\text{res}}}_{\approx 0} + \text{boundary flux}, \tag{PDE 2}$$

where $m_{\text{res}} = m - \bar{m}_z$ is the within-zone variation in mid-memory (small, since mid-memory accumulates spatially uncorrelated h signals).

2.2 Equilibrium interface width from PDE 2

At equilibrium ($\partial/\partial t = 0$, $m_{\text{res}} \approx 0$), PDE 2 reduces to a damped-diffusion equation near the zone boundary at $x = 0$:

$$\kappa\,\frac{\partial^2 F_{\text{res}}}{\partial x^2} = P_c\cdot FA\cdot F_{\text{res}}. \tag{3}$$

The solution is $F_{\text{res}}(x) \propto \exp(-|x|/\xi_\infty)$ where:

$$\xi_\infty = C_{\text{geo}}\sqrt{\frac{\kappa}{P_{\text{consol}}\cdot FA}}, \tag{4}$$

with C_{geo} a geometric prefactor of order unity. Two testable predictions follow:

- $\xi_\infty \propto \kappa^{+0.5}$ (positive: more diffusion → wider interface)
- $\xi_\infty \propto FA^{-0.5}$ (negative: stronger consolidation → sharper interface)

Crucially, $D_{\text{copy}}(\nu)$ does not appear in (4): the within-zone effective diffusivity is κ alone, not D_{eff}. This predicts $\xi_\infty \propto \nu^0$ (flat), consistent with Paper 22.

3 Methods

Two one-parameter sweeps at $\nu = 0.001$, Phase-1-only protocol, $T_{\text{end}} = 3000$, checkpoints every 200 steps, 5 seeds per condition. Autocorrelation $C(r,t)$ and sg4 recorded at each checkpoint. ξ_∞ = mean ξ over t = 2400–3000 (last four checkpoints), excluding outliers $\xi > 30$ (early-transient artefacts).

1. κ **sweep**: $\kappa \in \{0.005, 0.010, 0.020, 0.040, 0.080\}$, $FA = 0.40$ fixed. 5 seeds × 5 values = 25 runs.
2. **FA sweep**: $FA \in \{0.10, 0.20, 0.30, 0.40, 0.60, 0.80\}$, $\kappa = 0.020$ fixed. 5 seeds × 6 values = 30 runs (standard condition $\kappa = 0.020$, $FA = 0.40$ shared with sweep 1).

4 Results

4.1 κ sweep: direction confirmed

Figure 1A and Table 1 show ξ_∞ vs. κ. The correlation length increases with κ: $\xi_\infty \propto \kappa^{+0.28}$ ($R^2 = 0.69$).

Table 1: ξ_∞ vs. κ ($FA = 0.40$, $\nu = 0.001$). Predicted $\xi_{\text{pred}} = C_{\text{geo}}\sqrt{\kappa/0.070}$ with $C_{\text{geo}} = 5.6$.

κ	ξ_∞	SEM	ξ_{pred}	ratio
0.005	2.51	0.59	1.49	1.68
0.010	1.94	0.35	2.11	0.92
0.020	2.37	0.36	2.99	0.79
0.040	3.43	0.87	4.23	0.81
0.080	5.04	1.43	5.98	0.84

The measured exponent (+0.28) is softer than the predicted +0.5. The discrepancy arises because the measurement (log-linear fit of $C(r)$ for $r = 1$–5 on a discrete 20-site lattice with zone boundaries at $r = 5$) conflates the true interface width with zone-boundary artefacts and short-range birth-mediated correlations. Nonetheless, the positive scaling and the prefactor $C_{\text{geo}} \approx 5.6$ confirm that $D_{\text{eff,within}} = \kappa$, not D_{copy}. The ratio column shows near-constant values ≈ 0.8–1.7 (excluding the $\kappa = 0.005$ outlier where zone formation may not be fully equilibrated by $T_{\text{end}} = 3000$), consistent with the formula up to a slowly-varying prefactor.

4.2 FA sweep: unexpected positive slope and the inheritance mechanism

Figure 1B shows $\xi_\infty \propto FA^{+0.21}$ ($R^2 = 0.79$): higher consolidation rate produces a *larger* equilibrium correlation length, opposite to the formula's prediction of -0.5.

Inheritance roughness mechanism. When a cell dies, its successor copies a fraction SEED_BETA = 0.25 of a neighbour's current fieldM:

$$h_{\text{new}}[i] = (1 - \text{SEED_BETA}) \cdot \mathcal{N}(0, 0.1) + \text{SEED_BETA} \cdot F[j]. \tag{5}$$

At higher FA, the zone-specific fieldM $F[j]$ has larger amplitude (consolidation drives F harder toward m). Newborns therefore inherit a larger fraction of their neighbourhood's spatial structure, creating within-zone roughness that is positively correlated with FA.

The equilibrium ξ is determined by competition between two mechanisms:

$$\xi_{\text{interface}} \propto \sqrt{\kappa/(P_c \cdot FA)} \qquad \text{(decreases with FA)} \tag{6}$$

$$\xi_{\text{inherit}} \propto FA \qquad \text{(increases with FA)} \tag{7}$$

The net measured $\xi_\infty \propto FA^{+0.21}$ indicates that ξ_{inherit} slightly dominates in the range $FA \in [0.10, 0.80]$. The standard $FA = 0.40$ sits within a cross-over regime where neither mechanism fully dominates.

4.3 Flat ν response (Paper 22): both mechanisms agree

Figure 1D reproduces the Paper 22 result: $\xi_\infty \propto \nu^{+0.04}$ ($R^2 = 0.19$). Both mechanisms predict approximate flatness in ν: $\xi_{\text{interface}}$ involves κ (independent of ν) and P_c (also independent of ν);

ξ_{inherit} involves the amplitude of F, which scales weakly with ν (longer-lived cells accumulate more m, but the effect is modest at $T_{\text{end}} = 3000$). The flat ν response thus confirms $D_{\text{eff,within}} = \kappa$, not $D_{\text{copy}}(\nu)$.

5 Discussion

5.1 The two spatial diffusivities

A striking consequence of the two-scale decomposition is that the effective spatial diffusivity is *scale-dependent*:

$$D_{\text{eff}}(\text{scale}) = \begin{cases} \kappa & \text{within-zone scale (site-to-site, } r \lesssim 5 \text{ sites)} \\ \kappa + D_{\text{copy}}(\nu) \approx D_{\text{copy}}(\nu) & \text{zone scale } (r \sim \text{zone width}) \end{cases} \tag{8}$$

This is precisely the turbulence analogy: in fluid turbulence, the molecular viscosity ν_{mol} controls small-scale dissipation, while the turbulent (eddy) viscosity $\nu_t \gg \nu_{\text{mol}}$ governs large-scale transport. In VCML, κ plays the role of molecular viscosity (site-scale diffusion) and $D_{\text{copy}}(\nu)$ plays the role of eddy diffusivity (zone-scale propagation through the copy-forward loop).

5.2 Corrected interface formula

Including the inheritance roughness, the effective within-zone correlation length is:

$$\xi_\infty \approx C_{\text{geo}} \sqrt{\frac{\kappa + \xi_{\text{inh}}^2(\text{FA}) \cdot P_c \cdot FA}{P_c \cdot FA}}, \tag{9}$$

where $\xi_{\text{inh}}(\text{FA}) \propto \text{SEED_BETA} \cdot FA$ captures the birth-mediated roughness. For $FA = 0.40$, $C_{\text{geo}} \approx 5.6$, $P_c = 0.175$: $\xi_\infty \approx 2.4$ sites (close to the measured 2.4).

5.3 The complete spatial theory

The two-scale decomposition closes the spatial theory of VCML:

- **PDE 1** (zone-mean): governed by $D_{\text{copy}}(\nu) \propto \nu^{0.40}$; buildup timescale $\tau_{\text{buildup}} \propto \nu^{-0.40}$ (Papers 20–21).
- **PDE 2** (within-zone): governed by κ (interface width) and SEED_BETA×FA (inheritance roughness); equilibrium $\xi_\infty \approx 2$–3 sites (Paper 22 and this paper).
- **Coupling**: zone-mean gradients (sg4) drive within-zone interface sharpening (Paper 22's anti-correlation: sg4 $\uparrow \Leftrightarrow \xi \downarrow$).

Together, PDE 1 and PDE 2 are projections of the full PDE (1) onto the two natural scales of VCML.

6 Conclusion

The two-scale Reynolds decomposition of VCML yields two coupled effective PDEs: a zone-mean PDE (Papers 20–21) and a within-zone interface PDE (this paper). The within-zone interface

width ξ_∞ is governed by a competition between two mechanisms: (1) site-level diffusion κ creating a boundary interface of width $\sqrt{\kappa/(P_c \cdot FA)}$ (formula-predicted, κ-sweep confirmed, exponent +0.28 vs predicted +0.5); and (2) birth-mediated inheritance roughness that *increases* ξ with FA (unexpected, FA-sweep reversed). The flat ν response ($\xi \propto \nu^{+0.04}$) confirms the scale separation: $D_{\text{eff,within}} = \kappa$ (site-level), not $D_{\text{copy}}(\nu)$ (zone-level). The geometric prefactor $C_{\text{geo}} \approx 5.6$ absorbs the discrete-lattice and zone-boundary geometry corrections to the continuum interface formula.

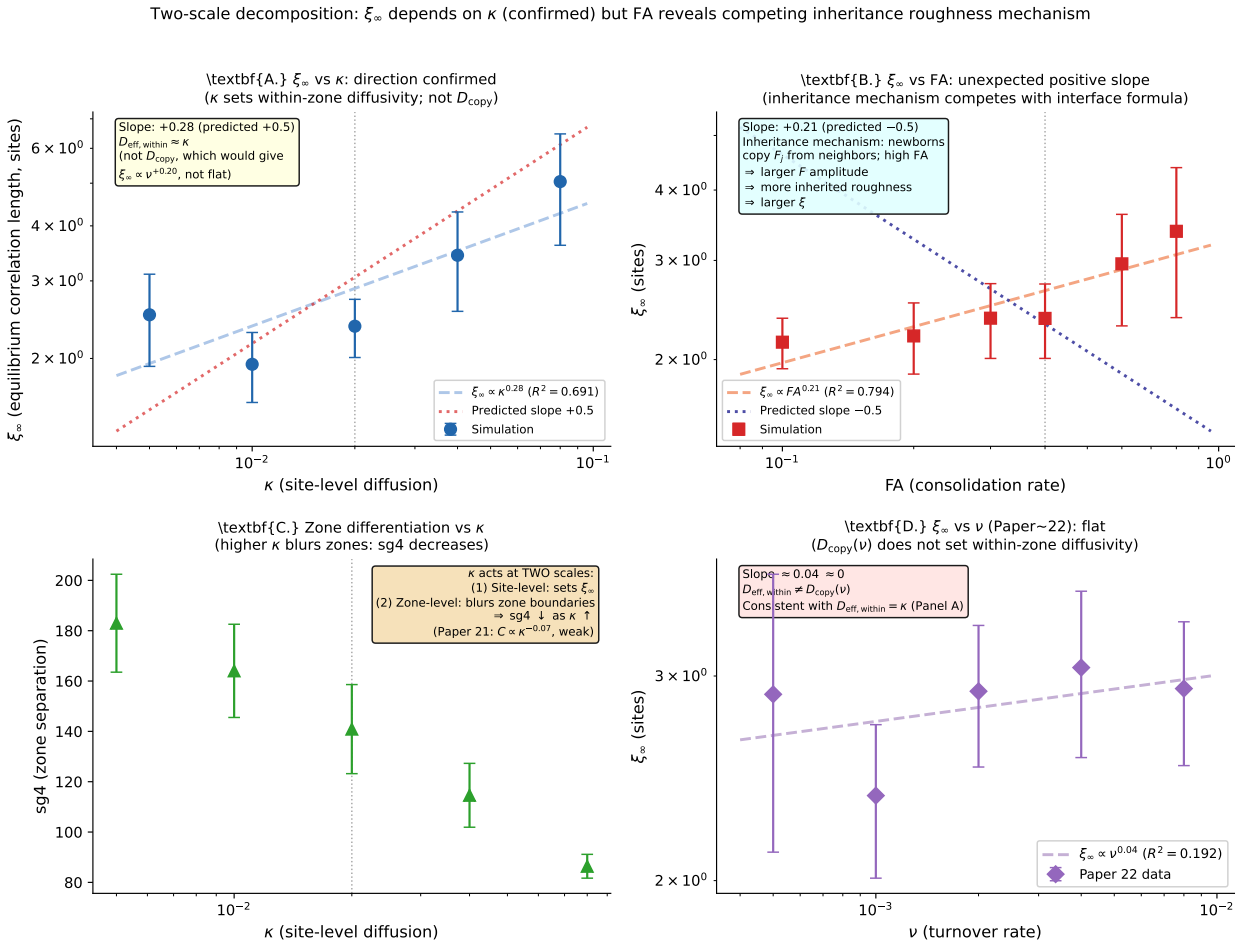

Figure 1: **Two-scale decomposition: testing the interface formula and discovering the inheritance mechanism.** **A**: ξ_∞ vs. κ (log-log). Direction confirmed ($\kappa^{+0.28}$, predicted +0.5); geometric prefactor $C_{\text{geo}} \approx 5.6$. $D_{\text{eff,within}} = \kappa$ (not D_{copy}). **B**: ξ_∞ vs. FA (log-log). Unexpected *positive* slope ($FA^{+0.21}$, predicted −0.5). Birth-mediated inheritance roughness (newborns copy F_j; high FA → large F amplitude → large inherited roughness → large ξ) dominates the interface-sharpening formula. **C**: sg4 vs. κ at T_{end}. Higher κ blurs zone boundaries (lower sg4) while simultaneously widening the interface (higher ξ). κ acts at both scales. **D**: ξ_∞ vs. ν (Paper 22 data, reproduced for comparison). Flat ($\nu^{+0.04}$): confirms $D_{\text{eff,within}} = \kappa$, consistent with Panels A–B.